Electric Vehicle Range Prediction Using Linear Regression

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Abstract--In today's world Electric Vehicles (EVs) are becoming increasingly popular due to environmental benefits also because the emission is less and the dependency on fossil fuels is less. One of the main concerns for EV users is that predicting accurately how far the vehicle can travel once fully charged. In this case, range prediction helps the drivers know how far their vehicle can go before the battery needs to be recharged. In this paper, there is a single method for predicting the range of an EV using the Linear Regression model. Linear Regression is a machine learning technique used to understand how a dependent variable changes based on one or more independent variables. Here factors like battery capacity, vehicle speed, energy consumption, and driving conditions are considered as input parameters for predicting the remaining travel distance.

Keywords—Electric Vehicle, linear regression, range prediction.

I. Introduction

Electric vehicles are gaining global attention due to increasing environmental concerns. Unlike other vehicles that run on fuel, EVs use electric motors with rechargeable batteries. However, the main challenge is to predict the vehicle's driving range accurately. Several factors like vehicle speed, battery capacity, driving style, and road conditions affect the drive range. This may affect the driver over the concern that the vehicle might run out of battery before reaching the destination. In order to overcome this issue, linear regression machine learning techniques are used. This paper proposes a linear regression-based model to predict the driving range of an electric vehicle using key parameters such as battery capacity and energy consumption.

II. Background of EV Range Prediction

Range prediction plays a critical role in EV energy management systems. The ability to estimate the remaining distance helps drivers plan their trips more effectively and prevents the battery from getting depleted during the journey.

Traditional range estimation methods often rely on fixed assumptions about energy consumption. However, in real-world driving conditions vary significantly and can affect the accuracy of such estimations.

Machine learning models analyze historical data to identify patterns and relationships between different variables that influence EV performance. Learning from this data, these models can provide more reliable and accurate predictions compared to traditional estimation methods.

Linear regression is particularly useful when the relationship between variables can be approximated as linear. This model learns from historical data and predicts the output variables based on the input parameters.

III. System Architecture

The EV range prediction systems consist of several stages including data acquisition, data preprocessing, model training, and range prediction.

A. Data Acquisition

The first step in the system involves collecting relevant data from the electric vehicle. The data can be obtained from the sensors and the battery management system. Important parameters include:

- Battery state of charge
- Vehicle speed
- Energy consumption rate
- Distance travelled
- Vehicle load

These parameters serve as input variables for the prediction model.

B. Data preprocessing

The second step in the system is data preprocessing, where the data is checked and cleaned, since it may contain missing values or noise. Hence, preprocessing is performed to improve data quality.

Data preprocessing includes:

- Removal of incomplete records
- Normalization of data values
- Filtering of noisy measurements

This step ensures that the dataset is suitable for training the machine learning model.

C. Model training

The next step after preprocessing, the dataset is divided into training and testing sets. The linear regression algorithm is then applied to the training data.

During the training phase, the model learns the relationship between the input parameters and the EV driving range. In this process, the algorithm calculates the coefficients that best fit the dataset to make accurate predictions.

D. Range prediction

Once the model is trained, it can predict the remaining driving range when the new input data is provided, i.e. when the vehicle is running. This prediction allows drivers to estimate how far the vehicle can travel before the battery needs to be recharged.

IV. MATHEMATICAL RANGE PREDICTION

The relation between dependent and independent variable must be done using linear equations in linear regression.

$$\mathbf{Y = a+bx}$$

Where,

Y = predicted range of EV

x = input variable

a = intercept of the regression line

b = slope coefficient

while we are using multiple variables the expression can be written as :

y = a + b1x1 + b2x2 + b3x3

here each parameter represents battery level, energy consumption and vehicle speed.

For reducing the error between predicted and actual value, the optimal value of coefficient is calculated in this model.

V. PARAMETERS INFLUENCING EV RANGE

There are few factors which affect the drive range of electric vehicle and they are listed below:

1). Battery Capacity: It determines the energy available to power the vehicle. The drive range increases when the battery capacity is large.

2). Vehicle Speed: As vehicle speed is increased, the aerodynamic resistance also increases, thus energy usage is increased.

3). Driving Behavior: Driving behaviors like harsh acceleration and applying brakes frequently increase energy consumption.

4). Vehicle Load: The net weight of the vehicle also plays a major role in energy consumption. If there is heavy load, then the energy consumption increases.

5). Environmental Factors: Different types of weather and climatic condition also influences the energy consumption.

These are the factors which affect the drive range of the vehicle.

VI. IMPLEMENTATION PROCEDURE

For implementing the EV range prediction, the following steps are followed:

a) Step 1: Collection of Data- EV sensors and battery management system(BMS) collects the driving data.

b) Step 2: Preparation of Data- The data set is cleaned and normalised to remove the inconsistencies if present.

c) Step 3:Training of Model- Using historical data, the model is trained in linear regression model.

d) Step 4: Testing of Model- The model is then tested using unknown data to check its performance.

e) Step 5:Range Prediction- Finally the model predicts the range by considering the input parameters.

VII. RESULT AND ANALYSIS

A. Using linear regression model and by selected parameters, the EV range prediction is done.

B. The results from the prediction is compared with the actual driving values to predict the model performance.

C. The results of the comparison shows that the model produces accurate prediction by training it with suitable historical data.

D. The predicted range matches with the EV drive range under a similar environmental condition which shows the efficiency of the model.

E. The complexity level of the model is relatively less which makes it applicable for all environment

F. Overall the EV range prediction model produced by analysing various data sets can be used effectively.

VIII. ADVANTAGES

i). Architectural Efficiency
The linear regression model maintains a low mathematical overhead when compared to high-dimensional neural networks. This simple approach is critical for EV applications as it allows the car's On-Board Electronic Control Unit (ECU) to process range estimates in real-time without draining significant batter power for computation.

ii). Deterministic Interpretability
Unlike the "black box" machine learning models, the linear coefficient in this study provides clear transparency regarding variable influence. This allows engineers to isolate exactly how much a single unit change in velocity or ambient temperature impacts the final range, facilitating easier model debugging and safety validation.

iii). Low-Latency Inference:
Once the model parameters are calibrated, the inference phase requires minimal floating-point operations for dynamic driving environments, enabling the system to update the range forecast instantly as sensor data fluctuates, rather than waiting for complex hidden layers to process.

iv). Resource-Light Deployment
Because the model avoids heavy memory-intensive libraries, it is highly scalable across low-cost hardware. This makes the solution commercially viable for entry-level EV models that may not have the high-end GPU or TPU acceleration required for deeper deep-learning architectures.

v). Enhanced Energy Management Integration
By providing a stable and predictable range output, this model integrates seamlessly with Energy Management Systems (EMS). It assists in optimizing battery discharges rates and route planning, effectively reducing range anxiety by offering a reliable safety margin for long-distance travel.

IX). LIMITATIONS

While the linear framework offers significant computational advantages, several inherent constraints must be acknowledged:

i). Linear Approximation of Stochastic Variables:
The model operates on the assumption of linearity between the input parameters and range. In practice, factors like aerodynamic drag and battery discharge curves are non-linear; thus, the model may experience increased error margins at extreme velocity or low State-of-Charge (SoC) levels.

ii). Sensitivity to Dataset Dimensionality:
The regressor's predictive fidelity is strictly bound by the granularity and variance of the training data. Small or "noisy" datasets lacking diverse driving cycles (e.g., urban vs highway) may result in reduced generalization when deployed in real-world scenarios.

iii). Omission of High-frequency Variables: To maintain a minimized resource footprint, the model excludes dynamic exogenous factors such as rolling resistance, instantaneous traffic density, and topographic elevation changes, which can cause significant fluctuations in real-time energy consumption.

iv). Predictive Ceiling Vs Non-Parametric Models:
There is an inherent accuracy trade-off when using linear techniques. While efficient, this model lacks the multi-dimensional mapping capabilities of Deep Neural Networks (DNNs), which can more effectively capture the subtle interactions between auxiliary loads and motor efficiency.

v). Steady-State Vs Transient Dynamics:
The model is optimized for steady-state conditions and may struggle to account for high-transient events, such as aggressive regenerative braking or steep gradient ascents, where energy flux is highly erratic.

X). FUTURE SCOPE

While the current linear framework provides a robust baseline for range estimation, future research will focus on integrating non-linear stochastic variables such as real-time traffic density and elevation profiles via GPS-linked topographic data. To improve the predictive fidelity without compromising the model's low computational overhead, we aim to explore Hybrid Ensemble techniques that combine linear efficiency with lightweight decision trees. Furthermore, the integration of Edge-IoT sensors could allow for a "Digital Twin" approach, where individual driver behavior and long-term battery health degradation are factored into the range algorithm in real-time. This would transition the system from a static estimator to a self-optimizing diagnostic tool within the vehicle's telematics unit.

XI). CONCLUSION

This study confirms that Linear Regression remains a robust, low-latency solution for real-time Ev range estimation. While complex neural networks offer higher theoretical accuracy, our approach prioritizes onboard feasibility by significantly reducing the computational load on the vehicle's ECU. By isolating the key variables-specifically battery Soc, velocity, ambient thermal conditions-the model provides a transparent framework that helps to mitigate the range of anxiety without requiring high-end processing hardware. Future iterations could further refine these predictions by incorporating non-linear battery aging factors, but for current primary-level energy management, this linear technique offers a highly scalable and implementable starting point.

XII). REFERNCES

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